

Professional Summary

Platform engineer with **10+ years** building resilient document-generation pipelines, layout engines, and developer-facing template systems. Specialised in high-throughput PDF rendering, semantic authoring DSLs, and turning brittle production-ops scripts into typed, snapshot-tested libraries that scale.

Technical Skills

- **Languages:** Java 21, Kotlin, Groovy, Python, SQL
- **Document & Print:** PDFBox, Apache POI (DOCX/XLSX), iText, PostScript, ICC colour profiles, font metrics
- **Layout engines:** Custom DSL design, semantic layout trees, pagination, snapshot testing, visual regression
- **Build & infrastructure:** Maven, Gradle, GitHub Actions, JitPack, Docker, JMH benchmarking
- **Testing:** JUnit 5, AssertJ, PDFBox-based PNG diff, layout-graph snapshots, mutation testing (Pitest)
- **Distribution:** Maven Central, Sonatype OSSRH, GPG signing, JitPack, semantic versioning discipline

Education

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>MSc Computer Science</b><br><i>University of Manchester</i><br>Distinction. Thesis: <i>Composable layout primitives for deterministic document rendering.</i> | 2019-2021 |
| <b>BSc Software Engineering</b><br><i>Imperial College London</i><br>First-class honours. Specialisation in compilers and static analysis.                       | 2015-2019 |
| <b>Oracle Java Certification</b><br><i>Professional track</i><br>Java 17 platform deep-dive: records, sealed types, pattern matching, virtual threads.           | 2023-2024 |

Projects

- **GraphCompose (Java 21, PDFBox, Maven, JMH)**  
Declarative Java PDF layout engine. Semantic DSL, slot-based templates, snapshot testing. Powers production CV / invoice / proposal pipelines for hiring tools and billing systems. (*Open source*)
- **Template Studio (Kotlin, Compose Desktop, PDFBox PNG diff)**  
Internal tool for evaluating CV, proposal, and invoice output across 14 design presets. PNG diffing, side-by-side layout, baseline freezing.
- **LayoutLint (Java 21, JavaParser, Spoon)**  
Static analyser that flags fragile authoring patterns (deeply nested rows, untyped offsets, implicit page breaks) before they ship to production.
- **ChromeForge (Java, GraphCompose, Pandoc bridge)**  
Editorial-magazine document toolkit built on GraphCompose: cinematic covers, pull quotes, multi-column flow, sidebar callouts.

Professional Experience

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| <b>Senior Platform Engineer</b><br><i>Northwind Systems</i><br>Led the reusable document-generation platform serving billing, hiring, and reporting flows across <b>8 product teams</b> . Reduced template maintenance time by <b>70%</b> by retiring per-team PDF scripts in favour of one canonical engine. | 2024-Present |
| <b>Software Engineer</b><br><i>BrightLeaf Labs</i><br>Built backend services and production document rendering pipelines processing <b>2M+ documents per month</b> . Drove the migration from iText to a custom layout engine, eliminating licensing risk and cutting p99 render latency from 1.4s to 380ms.  | 2021-2024    |
| <b>Backend Engineer</b><br><i>Helix Print Co</i><br>Maintained a high-volume invoice-printing service (15M PDFs/year) and authored the compliance test harness that gated every template change.                                                                                                              | 2019-2021    |

## Additional Information

**Languages:** English (Fluent), German (Intermediate), Spanish (Basic)

**Work Eligibility:** Eligible to work in the UK and the EU

**Open Source:** Maintainer of GraphCompose. Regular contributor to PDFBox issue triage.

**Speaking:** JVM Summit 2024, Devoxx UK 2025 — both on declarative document layout.